BCA (Sem-I) – Math (103)

(GE-1)

2021

Time : 3 hours

Full Marks : 40

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer ***all*** *sections as directed.*

Section–A

(Compulsory)

1. Pick up the correct alternative for each of the following questions : 10×1=10

(a) If $y = x^n$ then y_n is equal to :

(i) 0

(ii) n

(iii) $n!$

(iv) None of these

(b) If $y = e^{mx}$ then y_{n+1} is equal to :

(i) e^{mx}

(ii) $m^{n+1}e^{mx}$

(iii) m^n

(iv) None of these

(c) The minimum value of $\sin x$ in R is :

(i) -1

(ii) 1

(iii) 0

(iv) None of these

(d) $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ is equal to :

(i) $\frac{\pi}{2}$

(ii) $\frac{\pi}{4}$

(iii) π

(iv) None of these

(e) $\int_{0}^{\frac{\pi}{2}} \sin x dx$ is equal to :

(i) 1

(ii) -1

(iii) 0

(iv) None of these

(f) The area of the circle $x^2 + y^2 = 25$ is :

(i) 5

(ii) 25π

(iii) $5\pi^2$

(iv) None of these

(g) The number of arbitrary constants in the general solution of differential equation of third order is /are :

(i) 1

(ii) 2

(iii) 3

(iv) None of these

(h) Integrating factor of the differential equation $\frac{dy}{dx} + 2xy = 3$ is :

(i) x^2

(ii) $\frac{x^2}{2}$

(iii) $2x^2$

(iv) None of these

(i) The general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$ is :

(i) $\log y = kx$

(ii) $y = kx$

(iii) $xy = k$

(iv) $y = k \log x$

(j) The order and degree of the differential equation

$$\frac{d^2y}{dx^2}+5\left(\frac{dy}{dx}\right)^2+7y=\cos x \text{ respectively :}$$

(i) 2,2

(ii) 2,1

(iii) 1,2

(iv) None of these

Section–B

Answer any **three** questions : 3×5=15

2. State and prove Euler's theorem on homogeneous.

3. State and prove Taylor's theorem with Lagrange's form of remainder.

4. Find the maximum and minimum value of

$$2x^3-21x^2+36x-20$$

5. Explain the method of finding the solution of homogeneous differential equations.

6. Explain the method of $\sin x$ in Maclaurin's series.

7. Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

Section–C

Answer any **three** questions of the following :

3×5=15

8. If $y = \sin\left(m\sin^{-1}x\right)$, then prove that

$$\left(1-x^2\right)y_2 - xy_1 + m^2y = 0$$

9. If $u = \sin^{-1}\frac{x^2+y^2}{x+y}$, then show that

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$$

10. Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x}$ or $\int_0^{\frac{\pi}{2}} \log\cos x\,dx$

11. Apply Maclaurin's theorem to prove that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots$$

12. Solve any **one** of the following :

(a) $\frac{dy}{dx} + \tan x . y = \sec x$

(b) $xdy - ydx - \sqrt{x^2 - y^2}\,dx = 0$

13. Solve any **one** of the following :

(a) $\frac{d^2y}{dx^2} + a^2 . y = x \cos x$

(b) $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = x^3 e^{2x}$

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